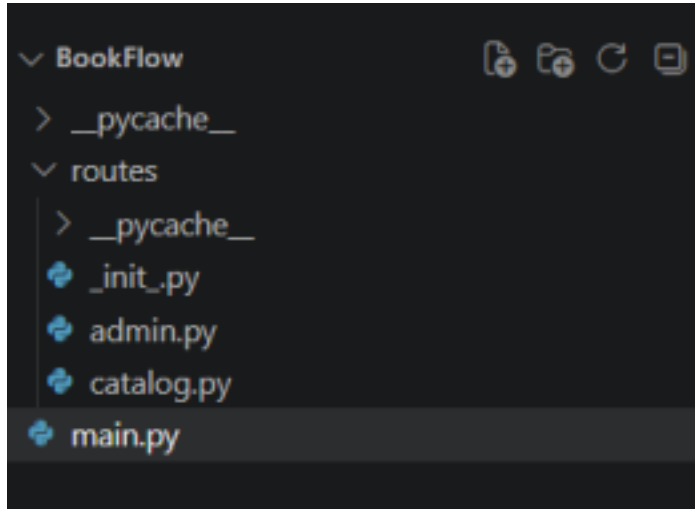


PRACTICAL WEEK - 6

Project Structure



catalog.py

```
routes > catalog.py > get_book
1  from fastapi import APIRouter, HTTPException
2
3  router = APIRouter()
4
5  # Temporary book data
6  books = [
7      {
8          "id": 101,
9          "title": "Python Programming",
10         "author": "John Smith",
11         "price": 450,
12         "isbn": "9781234567890",
13         "status": "Available"
14     },
15     {
16         "id": 102,
17         "title": "Database Management",
18         "author": "Robert Brown",
19         "price": 550,
20         "isbn": "9789876543210",
21         "status": "Available"
22     }
23 ]
24
25 # GET /books
26 @router.get("/books")
27 def get_books():
28     return books
29
30 # GET /books/{book_id}
31 @router.get("/books/{book_id}")
32 def get_book(book_id: int):
33     for book in books:
34         if book["id"] == book_id:
35             return book
36
37     raise HTTPException(
38         status_code=404,
39         detail="book not found"
40 )
```

```

router > admin.py > delete_book
1  from fastapi import APIRouter, HTTPException
2  from pydantic import BaseModel, Field
3  from routes.catalog import books
4
5  router = APIRouter()
6
7  # Pydantic model
8  class Book(BaseModel):
9      title: str
10     author: str
11     price: float = Field(ge=0)
12     isbn: str
13     status: str
14
15     # POST /admin/books
16     @router.post("/admin/books")
17     def add_book(book: Book):
18         new_id = max([b["id"] for b in books], default=100) + 1
19         new_book = {
20             "id": new_id,
21             **book.model_dump()
22         }
23         books.append(new_book)
24         return {
25             "message": "Book added successfully",
26             "book": new_book
27         }

```

```

28
29     # PUT /admin/books/{book_id}
30     @router.put("/admin/books/{book_id}")
31     def update_book(book_id: int, book: Book):
32         for index, existing_book in enumerate(books):
33             if existing_book["id"] == book_id:
34                 updated_book = {
35                     "id": book_id,
36                     **book.model_dump()
37                 }
38                 books[index] = updated_book
39                 return {
40                     "message": "Book updated successfully",
41                     "book": updated_book
42                 }
43             raise HTTPException(
44                 status_code=404,
45                 detail="Book not found"
46             )
47
48     # DELETE /admin/books/{book_id}
49     @router.delete("/admin/books/{book_id}")
50     def delete_book(book_id: int):
51         for index, book in enumerate(books):
52             if book["id"] == book_id:
53                 deleted_book = books.pop(index)
54                 return {
55                     "message": "Book deleted successfully",
56                     "book": deleted_book
57                 }
58             raise HTTPException(
59                 status_code=404,
60                 detail="Book not found"
61             )

```

```

main.py > ...
1  from fastapi import FastAPI
2  from routes.catalog import router as catalog_router
3  from routes.admin import router as admin_router
4
5  app = FastAPI(
6      |   title="BookFlow Library API"
7  )
8
9  app.include_router(catalog_router)
10 app.include_router(admin_router)

```

GET books

The screenshot shows a web browser window with the address bar displaying `GET /books`. The page has a light blue header with the text `GET /books` and a `Cancel` button. Below the header, there is a section for `Parameters` with the text `No parameters` and a `Cancel` button. A blue `Execute` button and a `Clear` button are visible. The `Responses` section shows a `Call` button and a `Request URL` field containing `https://127.0.0.1:8000/books`. The `Server response` section shows a `Code` field with `200` and a `Details` button. The `Response body` field contains a JSON array of two book objects:

```

[
  {
    "id": 1,
    "title": "Python Programming",
    "author": "John Smith",
    "price": 49.99,
    "isbn": "9781491560601",
    "status": "available"
  },
  {
    "id": 2,
    "title": "Database Management",
    "author": "Robert Brown",
    "price": 39.99,
    "isbn": "9781491560618",
    "status": "available"
  }
]

```

The `Response headers` section shows the following headers:

```

content-length: 288
content-type: application/json
date: Wed, 16 Sep 2020 09:16:45 GMT
server: uvicorn

```

GET /books/{book_id} Get Book

Parameters

Name

Description

book_id required
integer (path)

101

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8080/books/101' \
  -H 'accept: application/json'
```

Request URL

http://127.0.0.1:8080/books/101

Server response

Code	Details
200	Response body <pre>{ "id": 101, "title": "python programming", "author": "John Smith", "price": 499, "isbn": "9781111111111", "status": "Available" }</pre> Download Response headers <pre>content-length: 117 content-type: application/json date: Wed, 26 Sep 2024 09:17:04 GMT server: arkoon</pre>

Responses

Code	Description	Links
200	Successful Response	No info

POST books

POST /admin/books Add Book

Parameters

Cancel

Reset

No parameters

Request body required

application/json

Edit Value

Schema

```
{
  "title": "Artificial Intelligence",
  "author": "David Miller",
  "price": 499,
  "isbn": "9781111111111",
  "status": "Available"
}
```

Execute

Clear

Responses

Curl

```
curl -X 'POST' \
  https://127.0.0.1:8080/admin/books/ \
  -H 'content-type: application/json' \
  -d '{
    "title": "Artificial Intelligence",
    "author": "David Miller",
    "price": 499,
    "isbn": "9781120334444",
    "status": "available"
  }'
```

Request URL

https://127.0.0.1:8080/admin/books

Server response

Code	Details
200	Response body <pre>{ "message": "Book added successfully", "book": { "id": 10, "title": "Artificial Intelligence", "author": "David Miller", "price": 499, "isbn": "9781120334444", "status": "available" } }</pre> Response headers <pre>Content-Length: 120 Content-Type: application/json Date: Wed, 16 Sep 2020 09:13:13 GMT Server: Apache</pre>

Responses

Code	Description	Links
200	Successful Response	View Source

PUT books by ID

PUT /admin/books/{book_id} Update Book

Parameters

Cancel Reset

Name	Description
book_id * required integer (path)	933

Request body * required

application/json

Edit Value | Schema

```
{
  "title": "Advanced Artificial Intelligence",
  "author": "David Miller",
  "price": 599,
  "isbn": "9781120334444",
  "status": "available"
}
```

Execute Clear

Responses

Responses

Call

```
curl -X 'PUT' \
  http://127.0.0.1:3000/books/300 \
  -H 'accept: application/json' \
  -H 'content-type: application/json' \
  -d '{
    "title": "Advanced Artificial Intelligence",
    "author": "Herold Miller",
    "genre": "Sci-Fi",
    "isbn": "99922334455",
    "status": "Available"
  }'
```

Request URL

http://127.0.0.1:3000/books/300

Server response

Code	Details
200	Response body <pre>{ "message": "Book updated successfully", "book": { "id": 300, "title": "Advanced Artificial Intelligence", "author": "Herold Miller", "genre": "Sci-Fi", "isbn": "99922334455", "status": "Available" } }</pre> Response headers <pre>content-length: 368 content-type: application/json date: Wed, 16 Sep 2020 09:33:25 GMT server: uvicorn</pre>

Responses

Code	Description	Links
200	Successful Response	No links
422	Validation Error	No links

Media type

application/json

Controls Accept header:

Example Value | Schema

```
"string"
```

DELETE books by ID

DELETE

/admin/books/{book_id} Delete Book

Cancel

Parameters

Name	Description
book_id required	
Integer	100
(path)	

ExecuteClear

Responses

Curl

curl -X 'DELETE' \n'Mhttp://127.0.0.1:3000/admin/books/100' \n-H 'accept: application/json'

Request URL

http://127.0.0.1:3000/admin/books/100

Server response

Code	Details
200	Response body {\n "message": "Book deleted successfully",\n "book": {\n "id": 100,\n "title": "Advanced artificial intelligence",\n "author": "David Miller",\n "genre": "Sci-Fi",\n "isbn": "9781234567890",\n "status": "Available"\n }\n} Response headers content-length: 340\ncontent-type: application/json\ndate: Wed, 26 Sep 2025 09:22:51 GMT\nserver: uvicorn

Responses

Code	Description	Links
200	Successful Response	No links